

Notification Service smoke testing

Current scope: Week 1–3 .NET 8 API, SQL-backed jobs, RabbitMQ, and separate WorkerService. This guide supersedes the earlier recipient/body-logging guide. Operational logs must contain identifiers and outcomes, never recipient addresses, subjects, bodies, or credentials.

1. Preflight and local credentials

Run commands from the `notification-api` repository root. Docker Desktop must be using Linux containers. Ports 8080 and 15672 must be available on loopback.

```
dotnet --version
docker version
docker compose version
powershell -File scripts/Initialize-Week3Environment.ps1
docker compose -f compose.week3.yml config --quiet
```

The initializer creates an ignored `.env` and never prints its values. Compose fails closed if required credentials are absent. Do not paste the file into tickets, screenshots, logs, or evidence.

2. Build and automated verification

```
dotnet restore NSA.sln
dotnet build NSA.sln --configuration Release --no-restore
dotnet test NSA.sln --configuration Release --no-build --no-restore
dotnet ef migrations has-pending-model-changes --project NSA.csproj --configuration Release --no-build
dotnet list NSA.sln package --vulnerable --include-transitive
```

Pass condition: zero build warnings/errors, all tests pass, no pending EF model changes, and no vulnerable NuGet packages are reported.

3. Start and inspect the four-service stack

```
docker compose -f compose.week3.yml build api worker
docker compose -f compose.week3.yml up -d --force-recreate
docker compose -f compose.week3.yml ps
```

Wait until SQL Server, RabbitMQ, API, and worker all report `healthy`. Host access is deliberately loopback-only:

- Swagger UI: `http://127.0.0.1:8080/swagger`
- Liveness: `http://127.0.0.1:8080/health/live`
- Readiness: `http://127.0.0.1:8080/health/ready`
- RabbitMQ Management: `http://127.0.0.1:15672`

4. Week 1–2 HTTP contract smoke

Run the repository's repeatable 12-probe script after the API is healthy:

```
powershell -File scripts/Invoke-Week12Smoke.ps1 -BaseUrl http://127.0.0.1:8080
```

The helper verifies Swagger/OpenAPI, v1/v2 behavior, retirement headers, and representative RFC 7807 failures.

Area	Pass condition
Swagger/OpenAPI	UI and v1/v2 JSON return 200; documents report OpenAPI 3.0.x.
Versioning	v1 and v2 routes work; v1 includes <code>Deprecation</code> and <code>Sunset</code> ; v2 does not.
Errors	Business API errors return <code>application/problem+json</code> with status, title, instance, and trace ID.
CRUD	Create, list/read, update, and delete notification records; product/cart/order supporting probes succeed.

5. Week 3 asynchronous happy path

Use Swagger or an HTTP client to post a safe reserved-address example:

```
POST /api/v2/notifications/bulk
Content-Type: application/json

{
  "notifications": [{
    "recipientEmail": "visitor@example.test",
    "channel": 1,
    "subject": "Week 3 smoke",
    "body": "Non-sensitive training content",
    "orderId": null
  }]
}
```

Expect `202 Accepted`, a non-empty `jobId`, a `Location` header, and `x-Correlation-ID`. Poll the returned location until the persisted status is `Completed` or `CompletedWithErrors`. Verify `processedCount = succeededCount + failedCount`.

6. Recovery and DLQ proof

```
powershell -File scripts/Invoke-Week3RestartProof.ps1
powershell -File scripts/Invoke-Week3BrokerRestartProof.ps1
powershell -File scripts/Invoke-Week3InFlightBrokerRestartProof.ps1
powershell -File scripts/Invoke-Week3MalformedMessageProof.ps1 -MessageCase MalformedJson
powershell -File scripts/Invoke-Week3MalformedMessageProof.ps1 -MessageCase UnsupportedSchema
powershell -File scripts/Invoke-Week3MalformedMessageProof.ps1 -MessageCase WrongMessageType
powershell -File scripts/Invoke-Week3PoisonProof.ps1
```

Inspect the named DLQ `nsa.notifications.bulk.dlq` in RabbitMQ Management. Capture identifiers and counts only; redact credentials and message content. The poison helper enables failure injection only for its controlled run and restores the normal empty setting.

7. Health failure and recovery

When a dependency is deliberately stopped, process liveness must remain 200 while dependency readiness becomes 503. The worker must lose its healthy marker, then both services must recover after the dependency restarts. Use the recorded evidence procedure; do not delete volumes.

8. Stop safely

```
docker compose -f compose.week3.yml down
```

Do not add `--volumes` unless the owner explicitly intends to discard local SQL and RabbitMQ data. The generated credentials and named volumes are coupled; deleting or rotating only one side can make an existing volume inaccessible.

Evidence and boundaries

Store results under `evidence/week-01`, `evidence/week-02`, and `evidence/week-03`. The technical checks do not prove Skill IQ submissions, GitHub/PR publication, EM reviews, EM-supplied React snippets, Gist publication, or live presentation delivery.

Week 3 provides at-least-once delivery, not exactly once. Outbox, Inbox, HTTP idempotency, stable provider-side deduplication, Vault, CI/GHCR, and blue/green cutover remain later-week work.